

Linear Algebra II (2026 Fall)

Lecture 1.

Studying Contents : Preliminaries

© Review of Linear Algebra I

In this Linear Algebra II course, we will study

Chapters 5 (Eigenvalues and Eigenvectors) and 6 (Inner Product Spaces).

Therefore, the goal of today's lecture is to review, at a careful and gradual pace, the essential concepts from Linear Algebra I that are needed for these two chapters.

- vector spaces
- basis and dimension
- linear transformations
- matrices
- determinants
- invertibility

Question : In Calculus, we studied functions.

In Linear Algebra, what is the object that we study?

Answer : Vectors.

More precisely, vector spaces and linear transformations

This semester we shall study linear transformations much more deeply.

● Definition of Field

A field is defined to be a nonempty set F together with addition and multiplication such that for any $a, b, c \in F$ the following six properties hold :

- (1) $a + b = b + a$ and $ab = ba$
- (2) $a + (b + c) = (a + b) + c$ and $(ab)c = a(bc)$
- (3) There exist two different elements 0_F and 1_F in F such that $a + 0_F = a$ and $a \times 1_F = a$.
- (4) There exists an element $x \in F$ such that $a + x = 0_F$.
- (5) If $a \neq 0_F$, then there exists $y \in F$ such that $ay = 1_F$.
- (6) $a(b + c) = ab + ac$

Why do we need the concept of a field? The short answer is that, in order to work with scalar multiples of vectors, the scalars must support the four basic arithmetic operations: addition, subtraction, multiplication, and division (except by zero). This is exactly why we define a field.

This semester, we will not consider arbitrary fields. Instead, we will work exclusively with two fields: the field of real numbers, $\mathbb{R}$, and the field of complex numbers, $\mathbb{C}$.

● Definition of Vector Space

What is common among all vector spaces?

<u>Examples of Vector Spaces :</u>	$\mathbb{R}^n$	$\ni (a_1, \dots, a_n)$	vectors
	$P_n(F)$	$\ni a_0 + a_1x + \dots + a_nx^n$	polynomials
	$M_{m \times n}(F)$	$\ni A = (a_{ij})$	matrices
	$C[0,1]$	$\ni f: [0,1] \rightarrow \mathbb{R}$	continuous functions

These objects look completely different. Then why do we study them together?

Because **they satisfy exactly the same algebraic rules.**

A vector space over a field F is defined to be a nonempty set (of vectors) with
vector addition and scalar multiplication

which satisfies the following familiar algebraic rules :

(VS 1) $v + w = w + v$ for all $v, w \in V$

(VS 2) $(u + v) + w = u + (v + w)$ for all $u, v, w \in V$

(VS 3) There exists a vector $v_0 \in V$ such that $v_0 + v = v$ for all $v \in V$

(VS 4) For each $v \in V$, there exists a vector $w \in V$ such that $v + w = v_0$.

(VS 5) For each $v \in V$, it holds that $1 \cdot v = v$.

(VS 6) $(ab) \cdot v = a \cdot (bv)$ for all $a, b \in F$ and $v \in V$.

(VS 7) $a \cdot (v + w) = a \cdot v + a \cdot w$ for all $a \in F$ and $v, w \in V$.

(VS 8) $(a + b) \cdot v = a \cdot v + b \cdot v$ for all $a, b \in F$ and $v \in V$.

● Definition of Subspace

Definition : Let W be a subset of a vector space V over a field F .

We say that W is a subspace of V if

W is a **vector space over F**

with the operations **addition and scalar multiplication** defined on V .

Theorem 1.3. : Let W be a subset of a vector space V over a field F . Then

W is a subspace of $V \Leftrightarrow (i) \ 0_V \in W$;

(ii) $x + y \in W$ for any $x, y \in W$;

(iii) $cx \in W$ for any $c \in F$ and $x \in W$.

Example : Let F be a field and consider the vector space

$$\begin{aligned} P(F) &:= \text{the set of all polynomials in } x \text{ with coefficients in } F \\ &= \left\{ f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \mid n \geq 0, a_0, \dots, a_n \in F \right\}. \end{aligned}$$

Let $n \geq 1$ be a positive integer. Then

$$\begin{aligned} P_n(F) &:= \text{the set of all polynomials of degree } \leq n \text{ with coefficients in } F \\ &= \left\{ f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \mid n \geq 0, a_0, \dots, a_n \in F \right\} \end{aligned}$$

is a subset of $P(F)$.

Now, consider if $P_n(F)$ is a subspace of $P(F)$.

By Theorem 1.3, it is enough to check the following three conditions :

- (i) $0 \in P_n(F)$;
- (ii) $f + g \in P_n(F)$ for any $f, g \in P_n(F)$;
- (iii) $cf \in P_n(F)$ for any $c \in F$ and $f \in P_n(F)$.

So, $P_n(F)$ is **a subspace of** $P(F)$.

Example :

Some Terminologies : For an $m \times n$ matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$,

- a_{ii} 's are called the diagonal entries of the matrix A .
- $(a_{i1} \ a_{i2} \ \cdots \ a_{in})$ is called the i th row of the matrix A .

It is regarded as an element of the vector space $F^n = \{(a_1, \dots, a_n) \mid a_1, \dots, a_n \in F\}$.

- $\begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix}$ is called the j th column of the matrix A .

It is regarded as an element of the vector space $F^m = \left\{ \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \mid a_1, \dots, a_m \in F \right\}$.

- When $m = n$, we say that A is a square matrix.

Example (Continued) :

(1) An $n \times n$ matrix M is called a **diagonal matrix** if $M_{ij} = 0$ whenever $i \neq j$.

Thus M is of the form
$$M = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Let D_n be the set of all diagonal matrices in $M_{n \times n}(F)$. Then D_n is a subspace of $M_{n \times n}(F)$.

(2) Definition : A square matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in M_{n \times n}(F)$$

is called a **symmetric matrix** if $a_{ij} = a_{ji}$ for all (i, j) .

Now, let W be the set of all $n \times n$ symmetric matrices in $M_{n \times n}(F)$.

Then W is a subspace of $M_{n \times n}(F)$.

Example (Continued) :

(3) For an $n \times n$ matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in M_{n \times n}(F),$

we define the trace of A , denoted by $tr(M)$, is defined to be

the sum of the diagonal entries of A .

That is,

$$tr(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

The set $V := \{A \in M_{n \times n}(F) \mid tr(A) = 0\}$ is a subspace of $M_{n \times n}(F)$.

Theorem 1.4. : Let V be a vector space over a field F .

If $W_1, \dots, W_k$ are subspaces of V , then

the intersection $W_1 \cap \dots \cap W_k$ is also a subspace of V .

● Linear Combination

Definition : Let V be a vector space over a field F .

(1) Let $S = \{v_1, \dots, v_n\}$ be a subset of V .

A vector $v \in V$ is called a linear combination of vectors of S if

$$v = a_1v_1 + \dots + a_nv_n \text{ for some } a_1, \dots, a_n \in F.$$

In this case, we say that v is a linear combination of $v_1, \dots, v_n$ and

call $a_1, \dots, a_n$ the coefficients of the linear combination.

The set $\text{span}(S) :=$ the set consisting of all linear combinations of the vectors in S

$$= \{a_1v_1 + \dots + a_nv_n \mid a_1, \dots, a_n \in F\}$$

is called the span of S .

(2) Let S be a non-empty subset of V .

A vector $v \in V$ is called a linear combination of vectors of S if

$$v = a_1v_1 + \dots + a_nv_n \text{ for some } a_1, \dots, a_n \in F \text{ and } v_1, \dots, v_n \in S.$$

The set $\text{span}(S) :=$ the set consisting of all linear combinations of the vectors in S

$$= \{a_1v_1 + \dots + a_nv_n \mid n \geq 1, a_1, \dots, a_n \in F, v_1, \dots, v_n \in S\}$$

is called the span of S .

Theorem 1.5. : Let S be a non-empty subset of a vector space V .

(1) $\text{span}(S)$ is a subspace of V .

(2) $\text{span}(S)$ is the smallest subspace of V which contains S .

(This means that if W is a subspace of V , then $\text{span}(S) \subset W$ if and only if $S \subset W$.)

● Linear Independence

Definition : Let S be a subset of a vector space V . We say that S is linearly dependent if

there exist distinct vectors $u_1, \dots, u_n \in S$ such that

$$a_1 u_1 + \dots + a_n u_n = 0_V \text{ for some } a_1, \dots, a_n \in F, \text{ not all zero.}$$

In this case we also say that the vectors of S are linearly dependent.

Theorem 1.7. : Let S be a subset of a vector space V and let $v \in V$.

Suppose that S is linearly independent. Then

$$S \cup \{v\} \text{ is linearly dependent} \iff v \in \text{span}(S).$$

(Equivalently, $S \cup \{v\}$ is linearly independent $\iff v \notin \text{span}(S)$.)

● Basis of a vector space

Definition : Let V be a vector space over a field F . Then we say that

a subset β of V is a basis for V if (i) β is linearly independent and

(ii) β generates V (i.e., $V = \text{span}(\beta)$).

Example : In $V = F^n$, let $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $\dots$, $e_n = (0, \dots, 0, 1)$.

Then $\beta := \{e_1, \dots, e_n\}$ is a basis for V .

We call $\beta = \{e_1, \dots, e_n\}$ the standard basis for $V = F^n$.

Example : In $V = P(F) = \{a_0 + a_1x + \dots + a_nx^n \mid n \geq 0, a_0, \dots, a_n \in F\}$,

the set $\beta := \{1, x, x^2, \dots\}$ is a basis for $V = P(F)$.

Theorem : Let V be a vector space which has a finite basis, say $\mathcal{B} = \{v_1, \dots, v_n\}$.

Then every basis for V contains n vectors.

Definition : A vector space is called finite dimensional

if it has a basis consisting of a finite number of vectors.

The unique number of vectors in each basis for V is called the dimension of V

and is denoted by $\dim (V)$.

A vector space that is not finite-dimensional is called infinite-dimensional.

Question : (1) Explain why $\mathbb{R}^n$ as a vector space over $\mathbb{R}$ has dimension n .

(2) Explain why $\mathbb{C}^n$ as a vector space over $\mathbb{C}$ has dimension n .

(3) Explain why $\mathbb{C}^n$ as a vector space over $\mathbb{R}$ has dimension $2n$.

● Linear Transformations

Definition : Let V and W be two vector spaces over a field F .

(1) A map $T: V \rightarrow W$ is called **a linear transformation from V to W** if, for all $u, v \in V$ and $c \in F$, we have

$$(i) \quad T(u+v) = T(u) + T(v) \text{ and } (ii) \quad T(cv) = cT(v).$$

(2) Let $T: V \rightarrow W$ be a linear transformation.

- We define **the null space (or kernel) of T** , denoted by $N(T)$, to be

$$N(T) = \{ v \in V \mid T(v) = 0_W \}.$$

- We define **the range (or image) of T** , denoted by $R(T)$, to be

$$R(T) = \{ T(v) \mid v \in V \}.$$

Definition : Let $T: V \rightarrow W$ be a linear transformation such that

$N(T)$ and $R(T)$ are finite dimensional subspaces of V and W , respectively.

(1) The dimension of $N(T)$, denoted by $\text{nullity}(T)$, is called **the nullity of T** .

(2) The dimension of $R(T)$, denoted by $\text{rank}(T)$, is called **the rank of T** .

Theorem 2.3. (Dimension Theorem) : Let V be a finite dimensional vector space. If

$$T: V \rightarrow W$$

is a linear transformation, then the followings are true :

(i) $N(T)$ and $R(T)$ are finite dimensional vector spaces.

(ii) $\text{nullity}(T) + \text{rank}(T) = \dim V$.

● Matrix Representations

Definition and Remark : Let $T: V \rightarrow W$ be a linear transformation where

V and W are finite dimensional vector spaces over a field F .

Also let $\mathcal{B} = \{v_1, \dots, v_n\}$ and $\gamma = \{w_1, \dots, w_m\}$ be ordered bases for V and W , respectively.

(1) For each $1 \leq j \leq n$, we have

$$T(v_j) = a_{1j}w_1 + \dots + a_{mj}w_m \text{ for unique scalars } a_{1j}, \dots, a_{mj} \in F.$$

(2) The $m \times n$ matrix $[T]_{\mathcal{B}}^{\gamma}$ defined as

$$[T]_{\mathcal{B}}^{\gamma} := \begin{bmatrix} [T(v_1)]_{\gamma} & \dots & [T(v_n)]_{\gamma} \end{bmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

is called the matrix representation of T in the ordered bases $\mathcal{B}$ and γ .

(3) When $V = W$ and $\mathcal{B} = \gamma$, we write $[T]_{\mathcal{B}}^{\gamma}$ simply by $[T]_{\mathcal{B}}$.

Theorem : Let $T: V \rightarrow W$ be a linear transformation where

V and W are finite dimensional vector spaces over a field F .

Also let $\mathcal{B} = \{v_1, \dots, v_n\}$ and $\gamma = \{w_1, \dots, w_m\}$ be ordered bases for V and W , respectively.

Then

$$[T(v)]_\gamma = [T]_\gamma^\mathcal{B} [v]_\mathcal{B}.$$

Remark : Linear transformations are the starting point of Linear Algebra II.

In particular, Chapter 5 is devoted to

finding the best basis for representing a linear transformation.

In other words, Chapter 5 asks the following fundamental question:

Can we choose a basis so that

the matrix representation of a linear transformation becomes as simple as possible?

● Determinants

The determinant measures whether a square matrix is invertible. That is,

$$\det(A) \neq 0 \quad \Leftrightarrow \quad A \text{ is invertible.}$$

Definition 1. : Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. Then $\det(A)$ is defined to be $a_{11}a_{22} - a_{12}a_{21}$.

Definition 2. : For $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, we define $\det(A)$ by

$$\begin{aligned} \det(A) &= a_{11}\det\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12}\det\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{13}\det\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \end{aligned}$$

$$\det(A) = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$$

This formula says: Choose exactly one entry from each row and each column.

Each such choice corresponds to a permutation σ of $\{1,2,3\}$.

The product $a_{1,\sigma(1)}a_{2,\sigma(2)}a_{3,\sigma(3)}$ is one term of the determinant.

The sign is determined by whether the permutation is even or odd.

In Chapter 4, we learned several fundamental properties of determinants. These properties allow us to prove the following formula.

Leibniz Formula : Let $A = (a_{ij})$ be an $n \times n$ matrix. Then

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}.$$

Corollary : $\det(A - tI) = (-1)^n t^n + \text{lower terms}$

In particular, $\det(A - tI)$ is a polynomial of degree n .

Final Remark : In Linear Algebra I, the determinant was the end of the story.

We learned how to compute it and how to decide whether a matrix is invertible.

In Linear Algebra II, the determinant becomes the beginning of a new story.

Instead of evaluating $\det(A)$, we consider the polynomial $\det(A - tI)$,

whose roots reveal one of the most fundamental invariants of a linear transformation:

its eigenvalues.